

DAWSON COMMUNITY AP, MT, USA

WMO: 726676

Lat: 47.133N

Lon: 104.800W

Elev: 2457

StdP: 13.44

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 24087

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.1	-11.1	-25.5	1.5	-15.3	-18.5	2.2	-10.1	33.0	26.9	29.6	26.5	7.4	320	0.685

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	26.7	94.6	65.7	91.0	65.0	88.1	64.4	70.0	86.0	68.1	84.1	66.3	82.7	10.9	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.3	98.9	76.4	63.0	94.2	75.2	60.8	87.3	73.6	35.5	86.1	33.7	83.6	32.2	82.7	78.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
28.8	25.4	22.2	-24.3	100.6	6.3	3.2	-28.8	102.9	-32.5	104.8	-36.0	106.6	-40.5	108.9
			-23.4	73.4	4.3	2.4	-26.4	75.1	-28.9	76.6	-31.3	77.9	-34.4	79.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	44.1	16.8	20.0	32.2	43.2	53.3	63.3	72.1	69.9	59.5	44.9	31.8	20.1		
	DBStd	21.88	14.70	14.64	13.76	9.71	8.37	7.16	6.01	6.77	9.15	9.48	11.89	13.52		
	HDD50	4440	1030	840	559	239	62	2	0	0	22	210	550	926		
	HDD65	8172	1495	1259	1016	653	370	114	11	30	211	625	997	1391		
	CDD50	2274	0	0	7	36	164	402	684	618	308	52	3	0		
	CDD65	533	0	0	0	0	8	64	229	183	47	2	0	0		
	CDH74	7342	0	0	3	36	188	898	2944	2435	779	59	0	0		
	CDH80	3212	0	0	0	7	39	310	1384	1139	318	15	0	0		
	WSAvg	10.1	10.0	9.9	10.7	11.3	11.0	10.2	9.4	9.5	9.5	10.3	9.9	9.9		
	PrecAvg	14.2	0.4	0.4	0.5	1.3	2.4	2.7	2.2	1.2	1.2	1.2	0.5	0.4		
(15) Precipitation	PrecMax	19.6	1.2	1.2	2.2	3.7	6.6	5.3	6.8	2.8	4.9	3.5	2.3	1.0		
	PrecMin	9.6	0.0	0.0	0.0	0.4	0.6	1.0	0.4	0.1	0.2	0.4	0.0	0.0		
	PrecStd	2.5	0.3	0.3	0.5	0.9	1.5	1.3	1.6	0.7	1.1	0.8	0.6	0.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.6	56.0	71.7	79.3	84.3	93.4	99.3	99.2	93.6	81.3	65.6	51.9		
		MCWB	40.9	43.4	50.9	55.0	61.8	67.1	67.0	66.4	63.6	57.6	48.9	40.8		
	2%	DB	43.5	48.2	63.3	71.9	79.4	87.9	95.0	93.7	88.4	72.3	57.4	45.4		
		MCWB	36.8	39.6	47.2	52.4	59.5	65.9	66.5	64.8	62.1	53.8	44.3	37.3		
	5%	DB	39.1	43.3	56.8	65.7	74.6	82.6	91.3	90.5	82.3	65.8	52.4	40.8		
		MCWB	34.2	36.2	43.8	49.6	56.9	63.9	66.1	64.1	59.8	50.8	42.1	34.1		
	10%	DB	36.5	38.6	51.8	61.2	70.3	79.2	88.3	87.6	77.4	61.0	47.4	36.9		
		MCWB	32.4	33.7	41.6	46.7	54.9	61.8	65.4	63.8	57.9	48.2	39.4	32.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	41.1	44.5	51.5	56.8	65.2	71.3	73.8	70.8	67.3	59.2	50.2	41.4		
		MCDB	48.3	54.9	68.3	76.2	77.3	85.9	89.6	87.9	84.8	76.9	64.2	49.4		
	2%	WB	37.2	40.2	48.1	53.0	61.2	68.3	70.8	68.4	64.3	54.9	45.8	37.5		
		MCDB	43.2	47.8	62.0	70.1	75.3	82.6	87.0	85.9	81.1	68.4	57.2	44.7		
	5%	WB	34.7	36.8	44.8	50.2	58.6	65.8	68.8	66.6	61.9	51.8	42.5	34.9		
		MCDB	39.4	42.8	56.3	63.8	71.9	79.8	84.8	83.9	79.3	64.6	51.5	40.5		
	10%	WB	32.4	33.8	41.5	47.7	56.0	63.3	67.0	64.9	59.3	49.1	39.4	32.4		
		MCDB	36.4	38.9	50.2	59.8	68.0	76.7	83.2	82.3	76.0	59.8	46.4	36.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	18.8	19.7	22.1	23.6	23.1	23.7	26.7	27.3	26.4	22.6	20.8	18.4		
		MCDBR	20.3	24.4	31.6	33.1	30.4	30.4	32.4	33.2	34.8	32.2	28.5	22.1		
		MCWBR	15.3	16.8	18.5	17.8	15.4	14.0	12.3	12.9	15.5	17.1	17.7	15.7		
	5% WB	MCDBR	20.1	23.6	30.3	30.2	27.0	27.9	28.1	28.8	30.8	29.6	27.5	21.5		
		MCWBR	15.3	16.6	18.3	16.9	14.3	14.1	12.2	12.4	14.3	16.3	17.4	15.8		
(40) Clear-Sky Solar Irradiance	taub		0.250	0.257	0.274	0.309	0.331	0.348	0.355	0.359	0.319	0.288	0.260	0.245		
	taud		2.532	2.553	2.561	2.456	2.454	2.444	2.445	2.403	2.497	2.572	2.550	2.537		
	Ebn at Noon		277	298	306	301	296	290	286	281	285	280	270	266		
	Edh at Noon		19	23	27	33	34	35	34	34	28	23	18	17		
(44) All-Sky Solar Radiation	RadAvg		501	810	1195	1568	1861	2115	2257	1857	1371	876	546	390		
	RadStd		41	60	90	88	174	134	83	112	105	74	58	20		

Historical Trends

		DBAvg		Heating		Cooling		Degree-Days				
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (12 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page